# User Guide

## Zero-Inflated Poisson (ZIP) Latent-Class Trajectories

File: `zip_poisson.sas`    Traj2 (prototype, under QA)    Author: Anum Zafar

**Prototype.** This model is undergoing quality assurance. Interfaces and results may change before release.

## What it does

This file fits zero-inflated Poisson (ZIP) latent-class trajectory models for count outcomes with excess zeros, such as quarterly ED visits, in PROC NLMIXED. It sorts subjects into a few latent classes, each with its own count trajectory over time and its own probability of a structural zero. Everything is macro-only with no compiled code, so it runs as-is in the CMS VRDC and any SAS 9.4 or newer session. When counts are overdispersed (variance well above the mean), consider the ZINB model instead, which adds a dispersion parameter.

### What you need

- SAS 9.4 or newer with SAS/STAT (PROC NLMIXED).
- ODS Graphics for the plots (`%ct_zip_plots`). The fit itself runs without them.
- A wide, one-row-per-subject table of counts, and write access to WORK.

## The model in brief

Each class c has a mean count trajectory on the log scale and a zero-inflation probability that can also vary over time. The zero cell mixes a structural zero with a Poisson zero:

```
mean (log link):  mu_c(t) = exp(beta0_c + beta1_c*t + ...)
zero-inflation:   p_c(t)  = logistic(gamma0_c + gamma1_c*t + ...)

P(Y=0 | c) = p + (1-p) * exp(-mu)
P(Y=y | c) = (1-p) * Poisson(y; mu),   y > 0

class share:  w_c = exp(alpha0_c) / (1 + sum_k exp(alpha0_k))
```

Class A is the mixing reference (it has no alpha0). The zero cell and the mixing across classes both use a numerically stable log-sum-exp. Missing Y values are skipped in the likelihood, valid under missing-at-random.

## Quick start (simulated data)

The simulator creates toy counts; the input-prep block (section [B]) renames them to BASE_FILE_SRS with SUM_Q1..SUM_QT and BENE_ID. Set any starting values first (see below); unset ones start at 0.

```
%let T = 12;

/* 1. Simulate toy counts, then run section [B] to build BASE_FILE_SRS */
%sim_data(dist=poisson, n=5000, T=&T, seed=1);

/* 2. Fit a 3-class, quadratic-mean, constant-inflation model */
%ct_zip_pois_nlmixed(
  data=BASE_FILE_SRS, id=BENE_ID,
  yvars=SUM_Q1-SUM_Q12,
```

```
  nclass=3, class_labels=A B C,
  order=2, p_order=0, T=&T);

/* 3. Plot per-class curves + mixture mean */
%ct_zip_plots(pe=pe_zip, T=&T);
```

## The settings you edit

| Argument | Meaning |
|---|---|
| `data` / `id` | Input wide table and subject id (defaults `BASE_FILE_SRS`, `BENE_ID`). |
| `yvars` | Count outcome columns, one per time point, in time order (e.g. `SUM_Q1-SUM_Q12`). |
| `nclass` | Number of latent classes. |
| `class_labels` | Space-separated labels; the count must equal `nclass`. |
| `order` | Mean-curve degree: 1 linear, 2 quadratic, 3 cubic. |
| `p_order` | Zero-inflation degree: 0 constant, 1 linear, and so on. |
| `T` | Number of time points. |
| `tech` / `maxiter` | Optimizer technique (default `newrap`) and iteration cap (default 500). |
| `pe_out` / `fit_out` | Output dataset names for estimates and fit statistics. |
| `bounds` | Optional parameter bounds for stability. |

## Setting starting values

Starting values are set with %let lines in section [C] of the file, before you call the fit. Any parameter you do not set starts at 0. Class A has no alpha0 because it is the reference.

```
%let alpha0_B = -0.50;   /* mixing logits, class B onward */
%let beta0_A  =  0.10;   /* mean-curve coefficients: beta0/1/2/3_{class} */
%let beta1_A  =  0.02;
%let gamma0_A = -1.00;   /* zero-inflation logits: gamma0/1/2/3_{class} */
```

## Using your own data

Skip the simulator and section [B], and point data= at your own wide table with one row per subject. The fit does not read a separate time-index column: time is taken from the column order of yvars, so list the count columns in time order.

| Column | What to supply |
|---|---|
| Subject id | Named by `id`, unique per subject. |
| Counts | Named by `yvars`, one non-negative integer column per time point, in time order. |

- Counts must be non-negative integers. The excess zeros are what the zero-inflation part is for.
- Match `class_labels` to `nclass`, or the parameter build will not line up with the classes.

## Running and choosing the number of classes

Each call fits one model for a fixed nclass. To choose the number of classes, refit with different nclass values (and matching class_labels) and compare BIC in the fit-statistics dataset, preferring the lowest BIC with sensible class sizes. A commented ODS HTML5 block at the end of the file shows how to save the plots as PNGs.

## What you get back

- `pe_zip`: parameter estimates and standard errors (alpha, beta, gamma).
- `fit_zip`: fit statistics (AIC, BIC, -2 log L) for comparing candidate class counts.
- `traj` and `mix`: the per-class ZIP mean curves and the mixture-mean series.
- Two plots from `%ct_zip_plots`: per-class trajectories, and the same with the mixture mean overlaid.

## Troubleshooting

| Symptom | Fix |
|---|---|
| Errors about class parameters | class_labels count does not equal nclass. Make them match. |
| Will not converge | Set starting values in section [C], add bounds, or try fewer classes or a lower order. |
| Zero-inflation is flat over time | p_order=0 holds it constant. Raise p_order for a time trend in the structural zeros. |
| Counts are overdispersed | ZIP assumes variance equals the mean. If variance is much larger, use the ZINB model instead. |